

Multi-Objective Optimization by Quantum-Annealing-Inspired Algorithms

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Abstract

Combinatorial optimization is widely regarded as a primary application for near-term quantum processors, although a definitive demonstration of a practical quantum advantage remains elusive. Recent studies have reported that both gate-based quantum circuits and quantum annealers can outperform state-of-the-art classical heuristics on multi-objective optimization (MO-MaxCut) problems. However, these studies did not fully account for the substantial pre- and postprocessing overheads intrinsic to quantum solvers, leading to incomplete comparisons between quantum and classical approaches. In this work, we re-examine the same benchmark suite using GPU-based quantum-annealing-inspired algorithms (QAIAs), which generate probabilistic samples analogously to quantum processors and thus serve as formidable classical contenders. Our results show that QAIAs can sample candidate solutions approximately two orders of magnitude faster than previously studied quantum processors. In terms of end-to-end runtime, QAIAs also surpass industry-leading classical solvers, thereby providing the strongest performance among evaluated quantum and classical solvers for the MO-MaxCut instances considered here.

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Introduction

Combinatorial optimization is a major application domain for near-term quantum and quantum-inspired computing, and a central testbed for assessing whether such platforms can offer practical advantages on hard discrete problems [1]. This prominence is largely due to the fact that many industrial challenges—from logistics to financial modeling—can be mapped onto Ising or Quadratic Unconstrained Binary Optimization (QUBO) frameworks [2, 3], which are natively compatible with quantum architectures. In the current era of Noisy Intermediate-Scale Quantum (NISQ) technology [4, 5, 6], hybrid algorithms and specialized hardware aim to find high-quality solutions to NP-hard problems more efficiently than classical counterparts.

Within this domain, multi-objective optimization (MOO) [7, 8] addresses problems where multiple conflicting objectives must be optimized simultaneously. The goal of MOO is to identify Pareto-optimal solutions: trade-offs for which no objective can be improved without degrading at least one other. Their objective values form the Pareto front, the set of non-dominated points in objective space. Unlike single-objective optimization, MOO aims to approximate this front rather than return a single optimum. In this context, sampling-based solvers—such as gate-based quantum processors [9, 10, 11], quantum annealers [12, 13, 14], quantum-inspired optimizers [15, 16], and Ising machines [17]—are naturally relevant for the task. Indeed, unlike traditional iterative methods that isolate solutions sequentially, these samplers generate an ensemble of high-quality solutions, effectively mapping the Pareto front without the need for extensive computational cycles.

A preliminary study by Kotil et al. [18] investigated the application of the Quantum Approximate Optimization Algorithm (QAOA) [19, 20] to a specific class of MOO problems using IBM’s gate-based quantum hardware. Specifically, the authors demonstrated that gate-based systems could sample Pareto-optimal candidates for the three- and four-objective MO-MaxCut problem with 42 variables and hardware-native topology at speeds competitive with established classical algorithms.

Building upon this work, subsequent research by King [21] utilized Quantum Annealing (QA) [22, 23] via D-Wave’s Advantage series hardware to address the same problem instances. This follow-up study reported

that QA not only achieves higher sampling velocities but also requires a significantly lower sample volume to recover the complete Pareto set. As a result, QA can achieve the optimal hypervolume two to three orders of magnitude faster than the gate-based counterpart, highlighting its efficiency for these tasks.

However, these studies focused primarily on the time required to sample candidate solutions without accounting for the total execution time of the entire Pareto front generation process. A rigorous end-to-end comparison must encompass the model construction phase as well as the computationally intensive filtering of non-dominated solutions from the sampled set. Consequently, omitting these stages renders comparisons with classical baselines incomplete and may lead to an overestimation of the practical advantages offered by the quantum approaches.

In this work, we investigate the same problem instances using GPU-accelerated quantum-annealing-inspired algorithms (QAIA), focusing on Simulated Bifurcation (SB) [24, 25]. This framework enables us to evaluate modern classical Ising solvers on these benchmarks, with particular attention to the metrics critical for practical multi-objective optimization: sampling speed, sample diversity, and the efficiency of end-to-end Pareto-front recovery.

QAIA has previously demonstrated superior performance in solving single-objective MaxCut problems across diverse graph topologies [15, 16]. This motivates evaluating whether these algorithms exhibit similarly robust performance when extended to the multi-objective regime.

We show that GPU-accelerated QAIA provides a strong classical baseline for multi-objective quantum-optimization studies: it reaches the optimal hypervolume roughly one order of magnitude faster than QA and four orders of magnitude faster than QAOA. We further identify sample quality as a key factor in end-to-end efficiency, showing that QAIA recovers the complete Pareto set with about 10× fewer candidates than QAOA, though with a few times more samples than QA on average. To improve Pareto-set exploration, we introduce a Gaussian-noise variant of SB that increases candidate diversity. Finally, evaluating the full pipeline—model construction, sampling, and Pareto filtering—we show that the QAIA framework outperforms established Pareto-approximation heuristics, recovering the optimal Pareto front about one order of magnitude faster than the DCM and DPA-

a [26, 27] results reported in [18], while outperforming MOEA/D [28], NSGA-II [29], NSGA-III [30, 31, 32], and RVEA [33] in hypervolume and number of found Pareto-optimal solutions under fixed wall-clock time budget on dense three-objective instances up to $n = 200$.

These results indicate that, for the MO-MaxCut benchmarks considered here, GPU-accelerated quantum-inspired Ising solvers currently provide a stronger end-to-end baseline than the available quantum-hardware results. They also demonstrate the potential of approximate Ising solvers for complex MOO problems and provide a quantitative reference point for future advances in quantum optimization.

All code related to our work is publicly available on GitHub at <https://github.com/Tao-qubit/QAIA-for-MOO>.

Hamiltonian formulation of multi-objective MaxCut

Following Refs. [18, 21], we address the multi-objective weighted maximum-cut problem, which allows a direct comparison between QAIA and the methods reported in those studies on the same benchmark family. Consider an undirected graph $G = (V, E)$, where V is a finite vertex set, $E \subseteq \{(i, j) : i, j \in V, i < j\}$ is the edge set, and $n = |V|$.

For each of the K competing objectives, let $J_{ij}^{(k)}$ denote the weight of edge (i, j) in objective k . Using spin variables $s_i \in \{-1, 1\}$, the corresponding cut value can be written as

$$C_k(\mathbf{s}) = \frac{1}{2} \sum_{(i,j) \in E} J_{ij}^{(k)} (1 - s_i s_j). \quad (1)$$

Here, an edge contributes when its endpoints are assigned opposite signs. Since the first term is constant, maximizing $C_k(\mathbf{s})$ is equivalent to minimizing the Ising Hamiltonian

$$H_k(\mathbf{s}) = \sum_{(i,j) \in E} J_{ij}^{(k)} s_i s_j. \quad (2)$$

We therefore work directly with the Hamiltonian formulation, which is standard for Ising-based solvers [2].

The MO-MaxCut problem is formulated on the configuration space $\mathcal{S} = \{-1, 1\}^n$. Each configuration

$\mathbf{s} \in \mathcal{S}$ is assigned a vector of K Ising Hamiltonians,

$$\mathbf{H}(\mathbf{s}) = (H_1(\mathbf{s}), H_2(\mathbf{s}), \dots, H_K(\mathbf{s}))^T \in \mathbb{R}^K. \quad (3)$$

Since the objectives are minimized simultaneously, solutions are compared using the Pareto dominance order. A configuration $\mathbf{s}_1$ is said to dominate $\mathbf{s}_2$, denoted $\mathbf{s}_1 \prec \mathbf{s}_2$, if

$$\begin{aligned} H_i(\mathbf{s}_1) &\leq H_i(\mathbf{s}_2) \quad \forall i \in \{1, \dots, K\}, \\ \text{and } H_j(\mathbf{s}_1) &< H_j(\mathbf{s}_2) \quad \text{for some } j \in \{1, \dots, K\}. \end{aligned} \quad (4)$$

Equivalently, $\mathbf{H}(\mathbf{s}_1)$ Pareto-dominates $\mathbf{H}(\mathbf{s}_2)$ in objective space. The goal is therefore not to minimize a scalar objective, but to identify the set of Pareto-optimal configurations

$$\mathcal{P}^* = \{\mathbf{s} \in \mathcal{S} : \nexists \mathbf{s}' \in \mathcal{S} \text{ such that } \mathbf{s}' \prec \mathbf{s}\}. \quad (5)$$

The corresponding Pareto front $\mathcal{F}^*$ is the image of this set in objective space. In practice, the sampler produces a finite set $\hat{\mathcal{P}} \subseteq \mathcal{S}$ of nondominated spin configurations whose objective vectors approximate the Pareto front.

For evaluation, we report solution quality in the original cut-value space. Let $\mathbf{C}(\mathbf{s}) = (C_1(\mathbf{s}), \dots, C_K(\mathbf{s}))^T$. The quality of the obtained approximation $\hat{\mathcal{P}}$ is quantified using the *hypervolume* (HV) indicator [34, 35], defined as

$$HV(\hat{\mathcal{P}}, \mathbf{r}) = \Lambda \left(\bigcup_{\mathbf{s} \in \hat{\mathcal{P}}} [\mathbf{r}, \mathbf{C}(\mathbf{s})] \right), \quad (6)$$

where Λ denotes the K -dimensional Lebesgue measure and $\mathbf{r} \in \mathbb{R}^K$ is a reference point dominated by all objective vectors in the approximation set, i.e., $r_k \leq C_k(\mathbf{s})$ for all k and all $\mathbf{s} \in \hat{\mathcal{P}}$. $[\mathbf{r}, \mathbf{C}(\mathbf{s})] = \prod_{k=1}^K [r_k, C_k(\mathbf{s})]$ denotes the hyperrectangle spanned by the cut values of $\mathbf{s}$ and $\mathbf{r}$ (see Appendix A).

Scalarization via the weighted sum method

To solve the multi-objective problem using standard optimization solvers designed for single objectives, we employ the weighted sum method [8]. For a weight vector $\mathbf{c} = [c_1, c_2, \dots, c_K]^T$, where $c_k \geq 0$ and $\sum_{k=1}^K c_k =$

1, the scalarized Hamiltonian is given by

$$H_{\text{total}}(\mathbf{s}; \mathbf{c}) = \sum_{k=1}^K c_k H_k(\mathbf{s}) = \sum_{(i,j) \in E} J_{ij}(\mathbf{c}) s_i s_j, \quad (7)$$

where $J_{ij}(\mathbf{c}) = \sum_{k=1}^K c_k J_{ij}^{(k)}$ is the effective coupling for edge (i, j) . Consequently, the original multi-objective problem is transformed into a family of single-objective minimization problems:

$$\min_{\mathbf{s} \in \{-1, +1\}^n} H_{\text{total}}(\mathbf{s}; \mathbf{c}). \quad (8)$$

By solving these problems for different vectors $\mathbf{c}$, we obtain candidate solutions that approximate the Pareto frontier of the original multi-objective problem.

In the current work, we use a simplex-lattice design (Das–Dennis) [36] to generate weight vectors. Specifically, for K objectives, we generate a uniform lattice (resolution H) on the $(K-1)$ -dimensional unit simplex Δ^{K-1} , i.e., all nonnegative weight vectors summing to one with components restricted to $\{0, 1/H, \dots, 1\}$. This choice is well known in the area of multi-objective optimization [37]; it produces evenly spaced vectors and allows better reproducibility of the results.

Furthermore, we observed that the number of Pareto-optimal solutions obtained by an exact solver grows much slower than the number of vectors and seems to saturate well below 100 for the three-objective problem, while the full Pareto-set size is reported to be 2067 [21]. This indicates that the majority of the Pareto-optimal solutions are not global optima for any weight vector in the scalarized problem. We found that only a moderate number of well-spaced weight vectors is sufficient to allow the quantum-inspired solver to sample the Pareto set. In our experiments we also remove extreme weight vectors lying on the vertices and edges of the simplex and only use the subset corresponding to interior points. See more details in Appendix B.

Pareto filtering

For an Ising-based multi-objective solver, the sampler does not return a Pareto set directly. For a scalarization vector $\mathbf{c}$, the sampled Hamiltonian has the form introduced in Eq. (7), and repeated runs produce a finite pool of candidate spin configurations. The union of samples obtained over one or several scalarizations must therefore be converted into a nondominated ap-

proximation set by a classical postprocessing step.

Let

$$\mathcal{X} = \{\mathbf{s}^{(1)}, \dots, \mathbf{s}^{(M)}\}$$

denote the raw sample pool. Pareto filtering evaluates the objective vector $\mathbf{C}(\mathbf{s})$ for each $\mathbf{s} \in \mathcal{X}$, removes duplicate configurations or duplicate objective vectors if required, and retains only those configurations that are not dominated by any other sampled configuration. In cut-value maximization space, this yields

$$\hat{\mathcal{P}} = \{\mathbf{s} \in \mathcal{X} : \nexists \mathbf{s}' \in \mathcal{X} \text{ such that } \mathbf{s}' \prec \mathbf{s}\}.$$

The corresponding approximation front is then $\hat{\mathcal{F}} = \{\mathbf{C}(\mathbf{s}) : \mathbf{s} \in \hat{\mathcal{P}}\}$.

This filtering step is deterministic but can be computationally nontrivial, contributing significantly to the end-to-end runtime. A naive nondominated sorting procedure requires pairwise comparison of sampled configurations and has time complexity $O(KM^2)$ for M samples and K objectives. If many repeated configurations or repeated objective vectors are produced, deduplication reduces the effective input size; otherwise, filtering nondominated solutions becomes a computational bottleneck. For this reason, we treat Pareto filtering as a separate postprocessing stage in the computational pipeline and evaluate its overhead explicitly.

Simulated Bifurcation with Gaussian noise

To solve the composite Ising models constructed in the previous section, we employ the Simulated Bifurcation (SB) algorithm [24, 25]. For each weight vector $\mathbf{c}$, we run SB on the scalarized Ising matrix $J_{ij}(\mathbf{c})$; in the equations below, J_{ij} denotes this scalarized coupling matrix. SB is a heuristic algorithm inspired by the adiabatic evolution of quantum systems but implemented entirely on classical hardware by simulating classical Hamiltonian dynamics. It simulates a network of nonlinear oscillators that undergo bifurcations to converge toward low-energy states of the Ising Hamiltonian. In this work, we use two primary variants: ballistic SB (bSB) and discrete SB (dSB), and introduce a stochastic noise term to enhance sampling diversity for multi-objective optimization. A similar noise term is present in the SimCIM algorithm [38]. In SimCIM it is used to escape local minima and improve single-objective solution quality; here, we use it to increase sample diversity

Method	Type	Hardware information and details	Source
QA (Advantage)	Quantum	D-Wave Advantage System 4.1, measured access time	[21]
QA (Advantage2)	Quantum	D-Wave Advantage2 System 1.6, measured access time	[21]
QAOA (IBM_Fez) v1	Quantum	156-qubit IBM_Fez, $p = 6$, assuming 10,000 shots/s	[18]
QAOA (IBM_Fez) v2	Quantum	156-qubit IBM_Fez, $p = 6$, fidelity estimate 3.71% assuming 10,000 shots/s	[18]
QAOA (IBM_Fez) v3	Quantum	156-qubit IBM_Fez, $p = 6$, fidelity estimate 53.0% assuming 10,000 shots/s	[18]
QAOA (MPS- $\chi = 50$)	Classical	MPS simulation on Apple M1 Max CPU, bond dimension $\chi = 50$	[18]
QAOA (MPS- $\chi = 20$)	Classical	MPS simulation on Apple M1 Max CPU, bond dimension $\chi = 20$	[18]
DCM and DPA-a	Classical	AMD EPYC 7773X CPU	[18]
NI-bSB and NI-dSB	Classical	AMD EPYC 7773X CPU and NVIDIA GeForce RTX 4090 GPU	This work

Table 1: Hardware information for methods shown in Fig. 1.

for better coverage of the Pareto frontier. Overall, we found that SimCIM is less effective in this setting, see Appendix D.

In SB, each Ising spin $s_i \in \{-1, +1\}$ is represented by a continuous “soft spin” variable $x_i \in [-1, 1]$ and its conjugate momentum y_i . The system evolves according to the following ordinary differential equations:

$$\begin{aligned}\dot{x}_i &= a_0 y_i, \\ \dot{y}_i &= -[a_0 - a(t)]x_i + c_0 \left(\sum_{j=1}^n J_{ij} \phi(x_j) \right),\end{aligned}\quad (9)$$

where $a(t)$ increases linearly from 0 to $a_0 = 1$ over time, driving a bifurcation: initially all trajectories converge to the origin, but as $a(t)$ exceeds a critical threshold, the origin becomes unstable and x_i diverge toward ± 1 . Throughout this work, a_0 is fixed to 1 and the coupling scale is normalized as $c_0 = 1/\max_i |\sum_j J_{ij}|$. The function $\phi(x_j) = x_j$ defines bSB (ballistic SB), while $\phi(x_j) = \text{sgn}(x_j)$ defines dSB (discrete SB). The final spin assignment is $s_i = \text{sgn}(x_i)$.

To overcome the limited diversity of deterministic SB—which often converges to the same solution across runs for a fixed objective weight vector—we introduce Noise-Injected Simulated Bifurcation (NI-SB). We augment the momentum equation with Gaussian white noise:

$$\dot{y}_i = -[a_0 - a(t)]x_i + c_0 \left(\sum_{j=1}^n J_{ij} \phi(x_j) \right) + \alpha \eta_i, \quad (10)$$

where $\eta_i \sim \mathcal{N}(0, 1)$ is independently sampled for each spin at every integration step, and $\alpha > 0$ controls noise intensity. This perturbation acts as synthetic thermal noise, enabling trajectories to explore a diverse set of solution candidates. In the context of MOO, this mechanism significantly increases the probabil-

ity of discovering distinct Pareto-optimal solutions for the same weight vectors $\{\mathbf{c}^{(i)}\}$, thereby enriching the Pareto front approximation, as illustrated in Fig. 3(a).

Starting from small random initial values near the origin, the dynamical equations are integrated using an explicit Euler scheme with time step $dt = 1$. To enforce $x_i \in [-1, 1]$, an inelastic wall boundary condition is applied: whenever $|x_i| > 1$, we project $x_i \leftarrow \text{sgn}(x_i)$ and reset $y_i \leftarrow 0$, mimicking a perfectly inelastic collision.

Results and discussion

To evaluate the performance of the proposed Noise-Injected Simulated Bifurcation (NI-SB) framework, we conducted experiments on MO-MaxCut problems and compared the results against state-of-the-art quantum and classical solvers. The performance is assessed based on two key metrics: the speed of sampling candidate solutions for Pareto front approximation, and end-to-end performance of approximating the Pareto front measured by relative hypervolume and solution time.

Sampling time advantage

Fig. 1 illustrates the sampling performance of various algorithms. A full list of compared methods is presented in Table 1. The comparison includes QA executed on two generations of D-Wave Advantage systems, QAOA executed on IBM Fez QPU, MPS simulations of QAOA on classical hardware, classical exact algorithms (DCM, DPA-a), and our proposed NI-bSB/NI-dSB. The vertical axis represents the difference in hypervolume $(HV_{\max} - HV_t + 1)$, where a lower value indicates a better approximation of the true Pareto front. Following previous works, we inverted the axis for a more intuitive presentation.

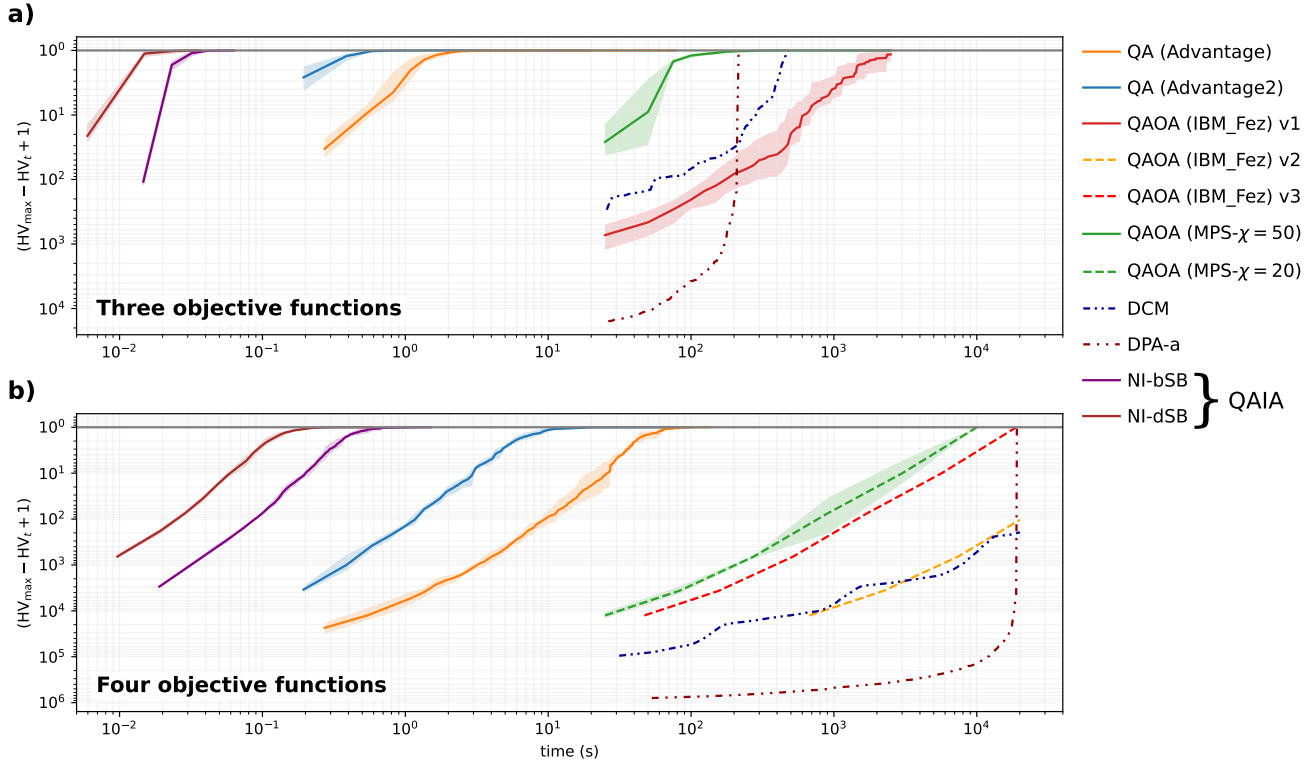


Figure 1: Convergence of different algorithms toward the Pareto front. The y-axis shows the Hypervolume difference ($HV_{\max} - HV_t + 1$) on a logarithmic scale, and the x-axis represents the sampling time in seconds. Data for all algorithms except SB are taken from Refs. [18, 21]. See the details and hardware information in Table 1.

We generated 190 evenly distributed Das–Dennis weight vectors for the three-objective problem and 220 for the four-objective problem. Each SB call was run with 50 iterations, a batch size of 3000 and a noise amplitude of 0.15 or 0.1 for the three- and four-objective problem, respectively. SB algorithms were called repeatedly until covering all Pareto-optimal solutions reported in previous works. On a GPU with 16 GB of memory, for a sparse problem with only 42 variables it is possible to sample all 3000 batches for all 220 weight vectors in parallel in a single run, using half-precision (float16) arithmetic.

As shown in Fig. 1, the NI-SB algorithms (purple and brown lines) exhibit a substantial advantage in speed of sampling the candidate set that contains the Pareto-optimal solutions. Both bSB and dSB reach the optimal hypervolume plateau within approximately 10^{-2} to 10^{-1} seconds. In contrast, QA systems require several seconds to minutes to achieve comparable performance, while QAOA and baseline classical methods are orders of magnitude slower. This result shows that leveraging GPU-accelerated classical dynamics allows for extremely rapid sampling of high-quality solutions,

making NI-SB highly suitable for time-critical applications.

Sample quality and end-to-end solution time

Although pure sampling-time efficiency is promising, it is the end-to-end time that defines practical algorithmic utility. This time includes not only the candidate-sampling time reported in [18, 21] and shown in Fig. 1, but also other sources of delay, including circuit compilation and parameter training for QAOA and—importantly—sample postprocessing to define nondominated solutions that constitute the Pareto front. Although all heuristic solvers deliver samples that contain Pareto-optimal solutions, postprocessing is still needed to define those solutions within the sample. This postprocessing time scales roughly linearly with sample size and may take a few seconds to a few minutes for millions of samples, as found for all three algorithms. This makes sample quality, i.e. the ratio of total sample size to the Pareto-set size, a crucial indicator of end-to-end performance.

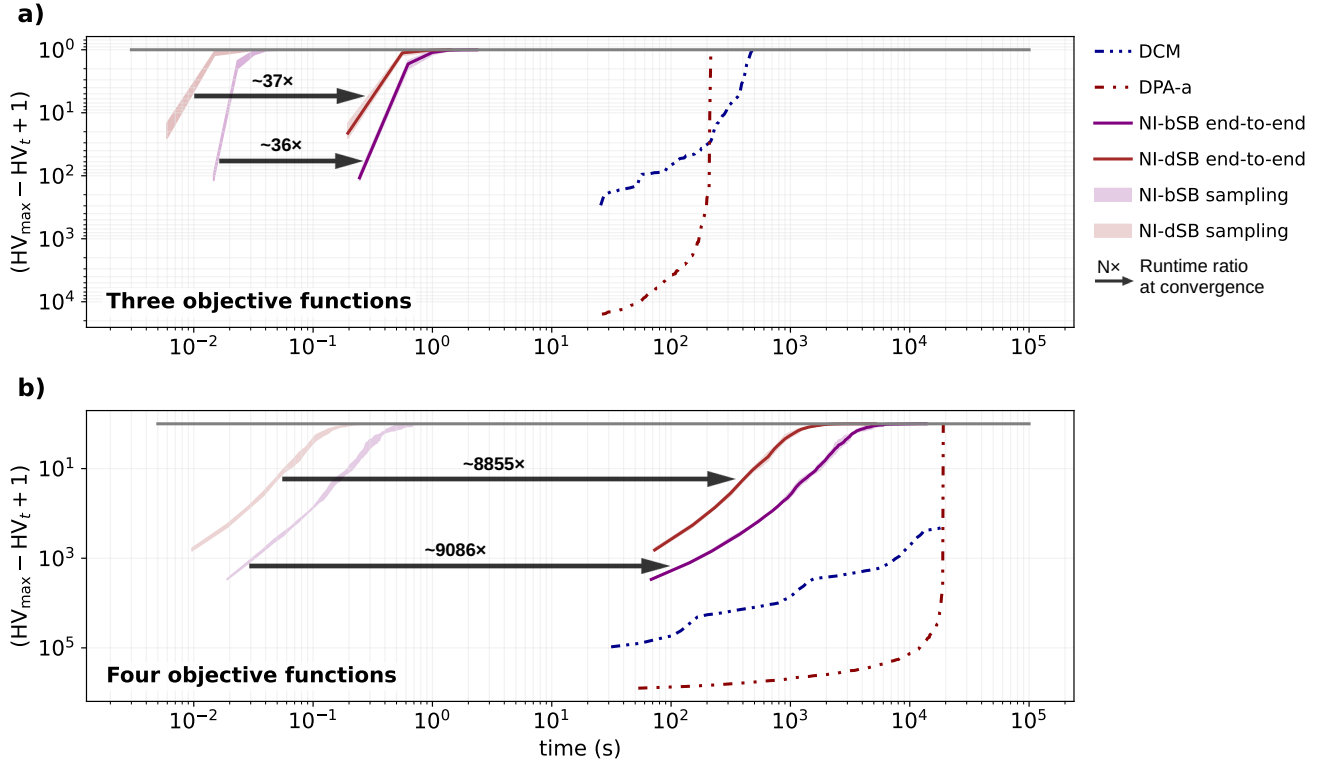


Figure 2: Comparison of sampling time with full Pareto front reconstruction time. The y-axis shows the Hypervolume difference ($HV_{\max} - HV_t + 1$) in logarithmic scale, and the x-axis represents the end-to-end time in seconds. DCM and DPA-a are classical baseline algorithms. End-to-end time for NI-dSB and NI-bSB includes both model construction and filtering of Pareto-optimal solutions from the sampled candidate set.

In Fig. 3 (b) and (c) we show that the total number of samples required by NI-SB to find the optimal Pareto set lies between the numbers reported in previous works for QAOA and QA. This suggests that sampling solutions with NI-SB should reduce postprocessing time compared to QAOA, while possibly remaining less sample-efficient than QA.

In Fig. 2 we show the end-to-end time taken by our NI-SB-based solver, including Ising model construction, sampling, and sample postprocessing to extract the Pareto set. We employ Fast Non-dominated Sorting for Pareto front filtering in the postprocessing stage, as implemented in the moocore Python package [39]. We compare this time to the execution time of classical baseline algorithms run on the same hardware. The results show that the QAIA-based solver is about an order of magnitude faster than those classical algorithms. Our analysis suggests that QA on moderate-size hardware-native graphs may show even better end-to-end performance.

Comparison to evolutionary algorithm-based heuristics

In the previous sections, we benchmarked our approach against various Ising solvers in terms of sampling speed and compared it end-to-end with classical exact algorithms. In this section, we further evaluate its performance against MOEA/D [28], NSGA-II [29], NSGA-III [30, 31, 32], and RVEA [33], as implemented in Py-moo [40]. These algorithms are well-established and widely adopted benchmarks in the multi-objective optimization field, representing distinct and influential methodological paradigms.

Drawing on the methodology in [41], we randomly generated three-objective MaxCut instances such that the cut value C_1 exhibits a strong negative correlation with C_2 , with Pearson’s coefficient ≈ -0.7 . The third objective C_3 is nearly uncorrelated with both C_1 and C_2 , with Pearson’s coefficient ≈ 0 . See more details on problem instance generation in Appendix E.

All algorithms are evaluated with a runtime limit of 5 seconds for instances with $n = 25, 42$, and 10 seconds

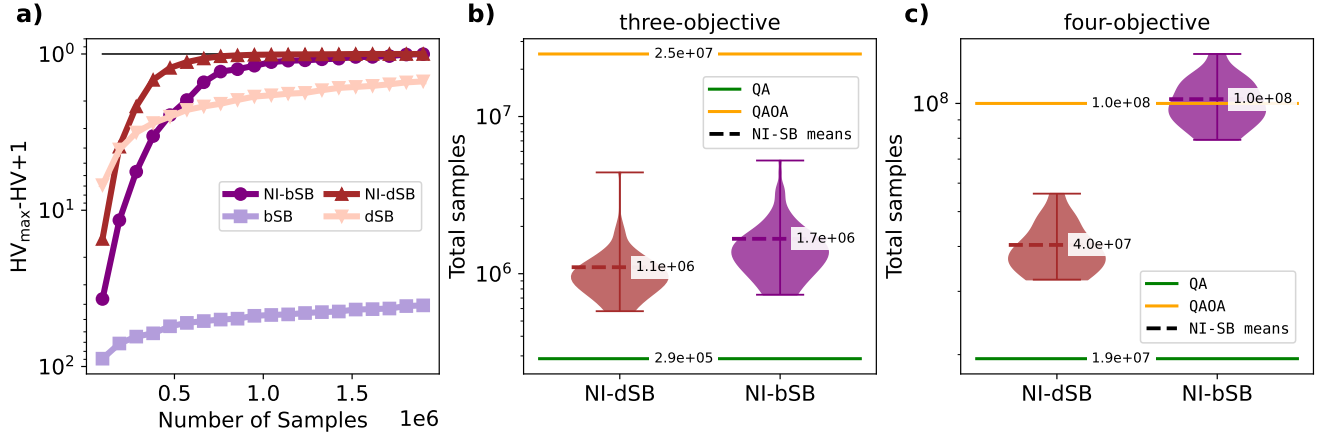


Figure 3: **a)** Convergence to optimal hypervolume for original and noisy SB. **b)** and **c)** Distribution of numbers of samples needed by NI-SB to discover the whole Pareto set of the three- and four-objective problems, respectively. Values for QA and QAOA are taken from [21] and [18], respectively. Note that QAOA result for four-objective problem is approximate, as it includes 10 nondominated solutions less than results of QA and NI-SB. Thus the plotted QAOA value in four-objective case should be interpreted as a lower bound on the number of samples required for complete front recovery.

otherwise. MOEA/D, NSGA-II, NSGA-III, and RVEA each use a population size of 190 and are terminated when reaching the runtime limit. The Das-Dennis method is used when reference directions are needed. Standard binary optimization settings are applied, including BinaryRandomSampling, TwoPointCrossover, and BitflipMutation, with GPU-accelerated function evaluations. Similarly, bSB and dSB utilize 190 Das-Dennis weight vectors and are configured with a noise amplitude of 0.1. For instances with $n = 25, 42$, the QAIA run for 50 iterations; otherwise, they run for 1000 iterations. Both algorithms generate batches of 570,000 candidate solutions in parallel for all weight vectors, followed by Pareto filtering. These sampling-filtering steps are repeated until the runtime limit is reached. A set of solutions filtered from randomly sampled candidate set is included as sanity baseline, subject to the same runtime limit.

We report both the number of recovered Pareto-front points and the hypervolume ratio achieved by each algorithm. Specifically, for instances with $n = 25$, the optimal Pareto fronts were obtained by brute force and the reference points for hypervolume calculation were determined by the minimum cut values of each objective. For the IBM data [18], the reference point and reference front were adopted from [18, 21]. For larger instances with $n = 100$ and $n = 200$, composite Pareto fronts were constructed by combining all non-dominated solutions found by the compared algorithms

and the reference points were estimated using the minimum values of each single objective observed across all obtained solutions. We generated 8 instances with varying sizes and densities. For each instance, every algorithm was executed for five independent runs for evaluation, and average values are presented in Table 2.

Brute-forcing an instance with a solution space of 2^{25} requires approximately 70 seconds. In contrast, within only 5 seconds, both QAIA and random sampling can generate roughly ten million samples and perform filtering, with the latter dominating the computational cost as shown in Fig. 2. Using this sample size, random sampling recovers over 80% of the optimal hypervolume, while QAIA captures the entire Pareto front. Evolutionary algorithms, despite producing significantly fewer solutions, still achieve more than 90% of the optimal hypervolume.

All experiments were conducted on a machine equipped with an AMD EPYC 7773X CPU and an NVIDIA GeForce RTX 4090 GPU.

The largest instance considered here is the complete graph on $n = 200$ vertices. This exceeds the reported 180 – 193 node estimate for the maximum clique size embeddable into Pegasus-16 and Zephyr-12 hardware graphs of the current D-Wave Advantage and Advantage2 quantum annealers [42, 43]. Even if these numbers improve as hardware progresses, it will require a significant increase in the number of qubits to leverage the advantage in parallel execution demonstrated

Instance				Random		RVEA		MOEA/D		NSGAII		NSGAIII		NI-bSB		NI-dSB	
n	d	Ref. front size	Reference type	Front points	HV Ratio	Front points	HV Ratio	Front points	HV Ratio	Front points	HV Ratio	Front points	HV Ratio	Front points	HV Ratio	Front points	HV Ratio
25	0.5	646	True	39	85.0%	26	86.7%	27	66.3%	116	94.8%	103	95.2%	646	100%	646	100%
25	1.0	649		42	84.2%	22	86.3%	22	64.5%	115	95.3%	78	94.9%	649	100%	649	100%
42	heavy-hex [18]	2067	Best-found [18, 21]	0	69.4%	12	92.2%	1	74.6%	45	95.6%	54	94.8%	2067	100%	2067	100%
100	0.5	4913		0	25.6%	1	57.8%	0	27.2%	4	62.4%	38	77.2%	2330	96.9%	3688	99.3%
100	1.0	4572	Composite	0	27.4%	5	62.5%	0	39.6%	3	58.1%	21	75.9%	2218	96.5%	3383	99.4%
200	0.5	3100	set	0	17.6%	10	60.6%	0	22.1%	13	60.9%	65	80.3%	1100	89.1%	2167	97.2%
200	1.0	1769		0	20.3%	24	72.4%	8	33.2%	25	63.7%	76	86.8%	551	81.1%	1183	92.4%

Table 2: Performance comparison of RVEA, MOEA/D, NSGA-II, NSGA-III, NI-bSB, and NI-dSB on three-objective MaxCut instances with varying node count (n) and edge density (d). Random uniform sampling is included to serve as the sanity baseline. For each algorithm, we report the mean number of recovered reference-front solutions and the mean achieved hypervolume ratio. All experiments were conducted on a machine equipped with an AMD EPYC 7773X CPU and an NVIDIA GeForce RTX 4090 GPU.

in Ref. [21].

Conclusions

We presented an efficient multi-objective optimization framework based on Quantum-Annealing-Inspired Algorithms (QAIA). The core of our approach uses a Gaussian-noise variant of Simulated Bifurcation (SB) to enhance sample diversity and mitigate premature convergence. We evaluated the method on MO-MaxCut instances previously used in quantum-computing studies.

Our results complement recent work on Ising-based approaches to multi-objective optimization [18, 21] by providing a strong (quantum-inspired) classical baseline. In numerical experiments, the proposed method generates high-quality samples substantially faster than QA as reported in [21], and recovers the complete Pareto set of three-objective MO-MaxCut with fewer candidate samples than QAOA, although it still requires more samples than QA on average.

In full end-to-end evaluations that include model construction, candidate sampling, and Pareto filtering, the method outperforms the considered classical heuristic baselines in hypervolume and number of recovered nondominated solutions under same time budget on synthetic MO-MaxCut instances. These findings suggest that performance advantages previously observed for QAIA in some single-objective settings also extend to multi-objective optimization.

On the benchmark instances from prior quantum studies, the full pipeline is faster than the DCM and DPA-a baselines reported in Ref. [18]. However, the advantage decreases as the number of objectives increases from three to four.

Although the available quantum-hardware results do

not yet match the end-to-end performance of GPU-accelerated QAIA on these benchmarks, they remain relevant because they belong to the same class of approximate Ising solvers. The higher sample efficiency reported for QA in [21] is particularly interesting and suggests that sample quality should be studied independently from sampling speed.

Future work should develop quantitative indicators that separate sampling speed from sample quality, such as nondominated yield, Pareto-set recall, diversity, and hypervolume gain per sample. It will also be important to test the approach beyond MO-MaxCut on a broader range of multi-objective optimization problems. Finally, other QAIA variants, such as thermally assisted SB [44], GSB [45], and Tabu-SB [46], should be examined for possible additional benefits.

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Use of generative AI

During the preparation of this work, the authors used generative AI tools, including Gwen, Kimi, ChatGPT, and NotebookLM, to support literature search, language editing, text drafting and refinement, code prototyping, and code improvement. Parts of the manuscript were also refined using the AI-enabled L^AT_EX editor Prism. These tools were used only as assistance in the preparation process and not as substi-

tutes for scientific judgment. All AI-assisted outputs, including text, code, references, data, analyses, and interpretations, were reviewed, verified, and revised as necessary by the authors. The authors take full responsibility for the accuracy, integrity, claims, results, conclusions, and overall content of this work.

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A Hypervolume indicator

The hypervolume (HV) indicator is a widely used scalar measure for comparing finite approximation fronts in multi-objective optimization. In this work, HV is computed in the original cut-value maximization space. Let $\hat{\mathcal{F}} \subset \mathbb{R}^K$ denote the set of nondominated objective vectors returned by the algorithm, and let $\mathbf{r} \in \mathbb{R}^K$ be a reference point dominated by all vectors in $\hat{\mathcal{F}}$, i.e., $r_k \leq y_k$ for all $\mathbf{y} \in \hat{\mathcal{F}}$ and all k . The hypervolume is the K -dimensional Lebesgue measure of the region dominated by $\hat{\mathcal{F}}$ and bounded by $\mathbf{r}$:

$$HV(\hat{\mathcal{F}}, \mathbf{r}) = \Lambda \left(\bigcup_{\mathbf{y} \in \hat{\mathcal{F}}} [\mathbf{r}, \mathbf{y}] \right), \quad [\mathbf{r}, \mathbf{y}] = \prod_{k=1}^K [r_k, y_k].$$

For a fixed reference point, HV is Pareto-compliant: if one approximation front dominates another, then it has a larger hypervolume. The indicator therefore combines information about convergence and spread of the approximation front, although its numerical value depends on the choice of reference point.

For illustration, consider a biobjective maximization problem with reference point $\mathbf{r} = (0, 0)$ and two nondominated objective vectors $\mathbf{y}_A = (10, 5)$ and $\mathbf{y}_B = (5, 10)$. The two dominated rectangles have areas 50 and 50, and their overlap has area 25. Hence,

$$HV(\{\mathbf{y}_A, \mathbf{y}_B\}, \mathbf{r}) = 50 + 50 - 25 = 75.$$

Adding a third nondominated vector $\mathbf{y}_C = (8, 8)$ increases the dominated area to 84, since it contributes the additional region $[5, 8] \times [5, 8]$.

B Das-Dennis simplex lattice and exact Pareto-optimal solutions

In Fig. 4 we show an example of a simplex lattice used to produce weight vectors for scalarizing the three-objective problem. We also solved corresponding single-objective problems to optimality using the exact MIP/MINLP solver SCIP [47], and show that the number of found Pareto-optimal solutions and the corresponding hypervolume grow much slower than the number of vectors in the lattice. This indicates that the majority of Pareto-optimal solutions do not correspond to an exact solution of the scalarized problem with any choice of weight vector. Thus an approximate algorithm such as NI-SB that samples a wide variety of near-optimal solutions is well suited for exploration of the Pareto set.

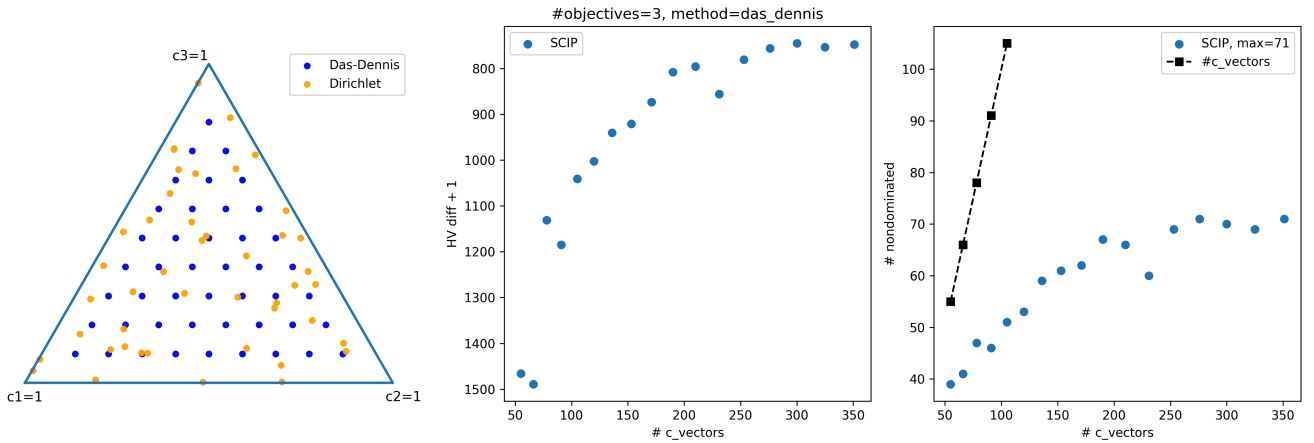


Figure 4: **Left:** Das-Dennis simplex lattice points compared with samples from a random Dirichlet distribution for three-objective problem. **Middle:** Hypervolume difference for solutions found by the exact solver as a function of the number of fixed weight vectors. **Right:** Number of Pareto-optimal solutions found by exact solver at each number of fixed weight vectors.

C Noise amplitude and effective sample size

In Fig. 5 we illustrate the choice of optimal noise amplitude α in Eq. (10) for both NI-bSB and NI-dSB. The figure shows that the main results of this paper (fast generation of samples that contain Pareto-optimal set, with the number of samples required to reproduce the whole Pareto front lying between the reported QA and QAOA results) hold over a wide range of noise amplitudes for the three-objective problem and are achievable with a properly chosen noise amplitude for the four-objective problem. The optimal value of α is found between 0.125 and 0.2 for both bSB and dSB in three-objective case, and around 0.1 for four-objective.

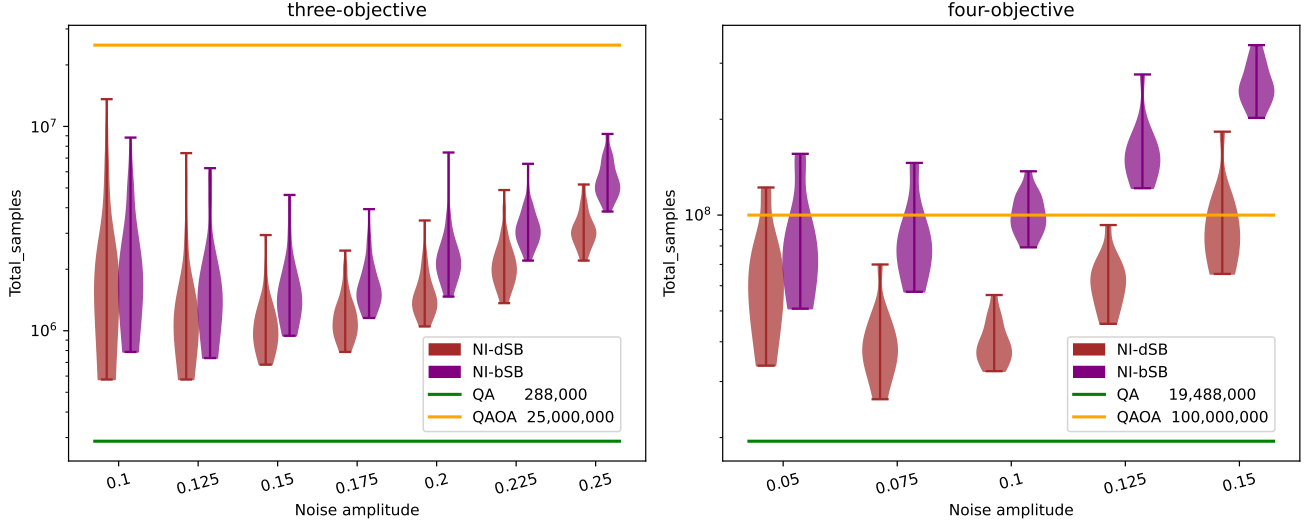


Figure 5: Total sample size required to reach optimal hypervolume, as a function of the noise amplitude injected in the algorithm. The QA and QAOA values are taken from Refs. [21, 18].

D Comparison with Simulated Coherent Ising Machine (SimCIM) and Simulated Annealing (SA)

Simulated Coherent Ising Machine (SimCIM) is a quantum-inspired optimization algorithm [38] that shares similarities with noise-injected simulated bifurcation methods. Both algorithms feature iterative update rules and incorporate noise terms in their formulations. We evaluate SimCIM and NI-SB on the three-objective problem from [18] using identical hardware. The results demonstrate that NI-SB achieves faster convergence than SimCIM. The key distinction lies in how momentum is introduced: in SB algorithms, momentum naturally emerges from the Hamiltonian equations and is set to zero when spin variables exceed the $[-1, 1]$ range, whereas in SimCIM momentum results from applying momentum gradient descent to the state-update equation without such boundary constraints. Our implementation of SimCIM utilizes MindSpore Quantum [48].

We also add results for Simulated Annealing, another common metaheuristic widely adopted for finding or approximating ground states of Ising Hamiltonians in the single-objective case. Execution of SA is inherently sequential, which makes GPU acceleration difficult, unlike SB and SimCIM. Thus we present results for SA executed on CPU (laptop AMD Ryzen AI 7 PRO 350), and add for reference results of NI-SB executed on the same machine. We used SA as implemented in D-Wave’s Neal library, set the number of sweeps and the batch size to values similar to those used for SB, and inverse temperature range $[0.1, 2]$.

This demonstration is not intended to provide an exhaustive comparison of the algorithms. We admit that SimCIM implementation in MindSpore Quantum may not be fully optimized to properly utilize GPU acceleration, as well as hyperparameters of both SimCIM and SA may be suboptimal. We selected relatively high temperatures

for SA by hand, and set the noise amplitude in SimCIM to the same value with SB. This experiment demonstrates why we choose SB as the primary candidate for our work. Comprehensive comparison, including other variants of QAIA may be the subject of future work.

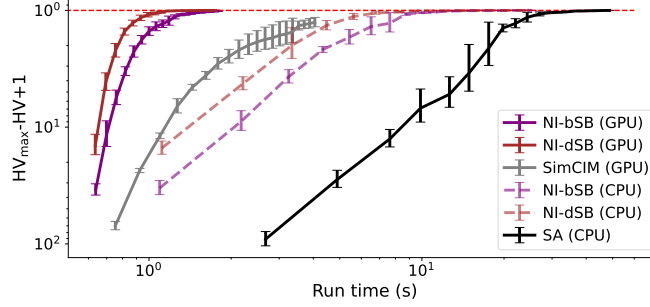


Figure 6: Comparison of SA, SimCIM and NI-SB. The y-axis shows the Hypervolume difference ($HV_{\max} - HV_t + 1$) on a logarithmic scale, and the x-axis represents the end-to-end run time in seconds.

E Solution Distribution and Pareto Front of MO-MaxCut Instances

The MO-MaxCut instances for the experiment in Table 2 are generated randomly. Specifically, we first generate two $n \times n$ matrices, A and B , with integer elements uniformly sampled from the interval $[-25, 25]$. We then construct two new matrices, $J^{(1)}$ and $J^{(2)}$, via a linear transformation:

$$\begin{pmatrix} J^{(1)} \\ J^{(2)} \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} A \\ B \end{pmatrix}, \quad (11)$$

where the parameters are set to $a = 1, b = 1, c = 0.2, d = -5$. This configuration results in a negative correlation between the cut values C_1 and C_2 , with a Pearson correlation coefficient of approximately -0.7 . Here, C_i is defined by Eq. 1. The third objective matrix, $J^{(3)}$, is generated independently using the same method as A and B . Additionally, a random mask matrix is applied to control the edge density of the graph. Figure 7 illustrates the solution distribution for a random MO-MaxCut instance with $n = 10$ variables.

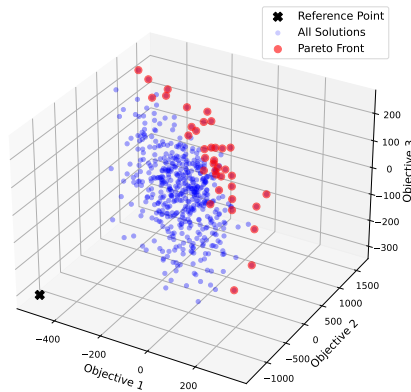


Figure 7: Solution space of a random MO-MaxCut instance with $n = 10$ variables. The blue points represent all feasible solutions, while the red points highlight the Pareto front.